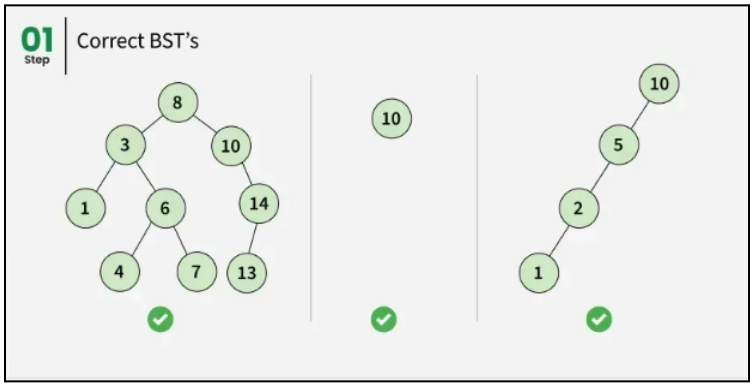
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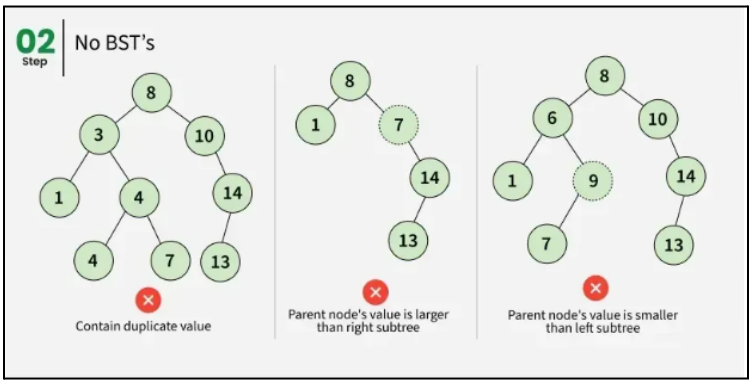
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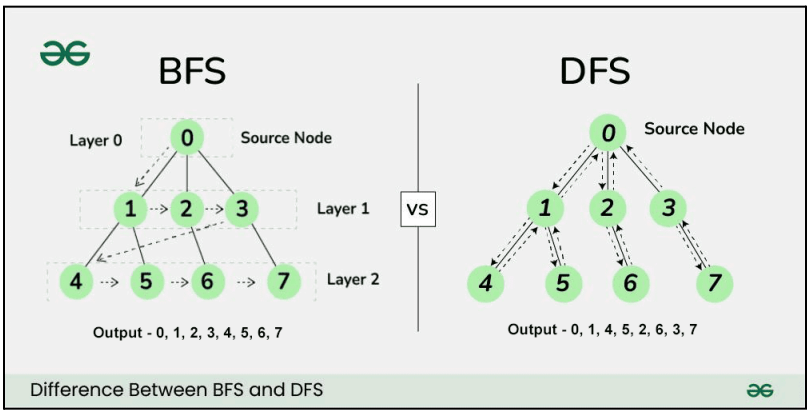
COURSE CODE/TITLE:	CPE 010 | Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	September 15, 2026	**DATE SUBMITTED:**	September 15, 2026
NAMES:	**Ramos**, Samuel **Reyes**, Fenzie Aaron **Rodulfa**, Jose Martin **Salcedo**, Lance Christoper **Santiago**, Niña Denise	**INSTRUCTOR:**	Engr. Jimlord Quejado

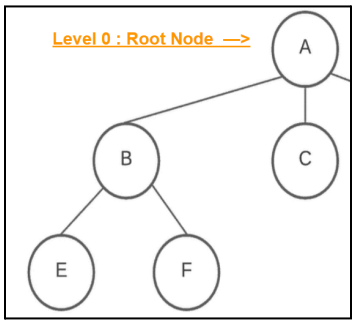
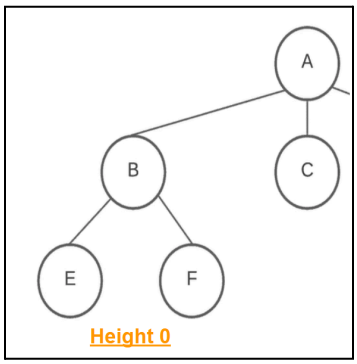
ASSIGNMENT NO._9.1_ **Binary Search Trees**

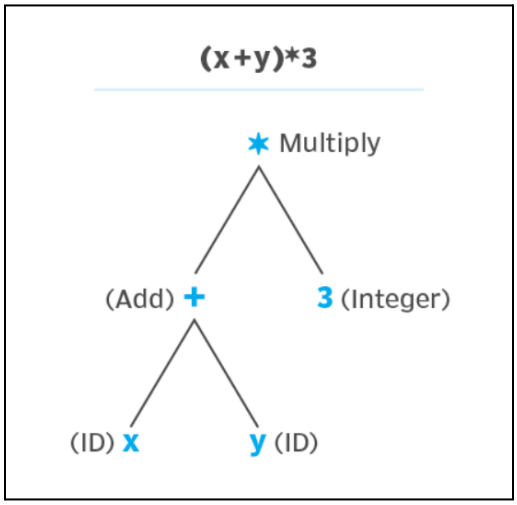
1. PROCEDURE OUTPUT

Questions:	**Answers:**
1. What is a binary search tree?	A binary search tree is a type of binary tree data structure in which each node has at least two children and also has a strict ordering property. For the left subtree, it contains only nodes that have strictly less than the parent node's value and for the right sub tree, the nodes must contain a strictly greater than the parent node's value. Correct:  *In this example the binary search is correct since it has no duplicates, the left subtree doesn't have a larger value than the parent node's value and the single node 10 didn't violate any of the rules.*

	Incorrect:  *In this example, the BST is incorrect since it contains a duplicate value (in this case 4), the parents node's value is larger than the right subtree (the parent node's value is 8) and the parent's node value is smaller than the left subtree (left side has 9 in the subtree but the parent node's value is 8).* Reference: GeeksforGeeks. (2026, February 16). *Binary search tree*. GeeksforGeeks. https://www.geeksforgeeks.org/dsa/binary-search-tree-data-structure/
2. Where can binary search trees be used?	When we deal with a large amount of information, it is crucial to have a way to arrange the data so that it can be accessed, organized, and be managed more efficiently. This is where the Binary Search Trees take place, as it can be used in situations where data needs to be searched, added, or removed frequently, such as in multilevel indexing in databases, dynamic sorting, managing virtual memory areas, file manager/file organization, dictionaries, and search systems. Since the values are arranged according to their order, the tree can decide whether to search on the left or right side instead of manually checking each and every element one by one. This makes searching and managing data more organized and efficient, especially when the tree is balanced. (REKA, et al., 2025) **Reference:** REKA, S., Vidyashree, V., Vishwas, S., Wahid, M., Yashas, C. S., Yashwanth Gowda K. S., Yashwanth, V., Niveditha, H., & Rahul Mukesh, R. (2025). *Binary search tree overview and applications*. *Scribd*. *https://www.scribd.com/document/866420063/Fdt-Report-Word*

3. What is a tree traversal?	Tree traversal is the process of passing or going through every node of the binary tree exactly once in a specific order to read its data. Since trees are hierarchical, there are multiple ways to visit every note of the binary tree, such as Depth-First-Search which visits nodes by going as deep as possible down a branch before backtracking, and Breadth-First Search which visits nodes level-by-level from top to bottom and left to right within each level of the binary tree.  **Image from:** BFS vs DFS for Binary Tree - GeeksforGeeks

4. Differentiate a post-order and pre-order traversal give examples.	Pre-order and Post order Traversal are two different ways to access the nodes of a tree. In pre-order traversal, the root node is visited first, then proceeds to the left subtree and followed by the right subtree. In simple terms, it starts its accessing at Level 0 down to level (n).  *In this example, with pre-order traversal the order would be accessing root node A first, then left child B, and right child C. So the order will be A - B - C.* On the other hand, post-order traversal visits the left subtree first, then the right subtree, and last is the root node. In simple terms, it starts its access at Height 0 up to Height (n).  *In the same example, with post-order traversal the order would be accessing left subtree (B) first, then right subtree (C), and last root node (A). So the order will be B - C - A.* **Image from:** Dev Community

5. What is a parse tree and its use?	A parse tree is essentially a properly sorted, hierarchical tree that contains the syntax of a code/a sentence. It breaks down a source code, sorts each syntax of the code depending on priority and hierarchy, and is visualized via a tree.  **Image from:** TechTarget *An example of the usage of a parse tree, where we visualize the hierarchy of syntax that a compiler reads, as well as the logic of a compiler if it were to solve the given problem.* As shown in the figure above, parse trees are useful in understanding compiler logic and systematically breaking down lines of code in order to catch errors, visualize syntax hierarchy, and interpret source code.

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2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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